big data pipeline ML analysis

Flight Delay and Cancellation Prediction System

Team 14

Vasilev Ivan
Sarantsev Stepan
Sharafutdinov Farit
Belyaev Grigorii

Goal

Flight Cancellations cost everyone

Goal

Flight Cancellations cost everyone

$8B+

Annual revenue lost
by U.S. airlines

Goal

Flight Cancellations cost everyone

$8B+

Annual revenue lost by U.S. airlines

~2-5%

of all flights cancelled (up to 5.6% for worst carriers)

Goal

Flight Cancellations cost everyone

$8B+

Annual revenue lost by U.S. airlines

~2-5%

of all flights cancelled (up to 5.6% for worst carriers)

25 min

Average delay per affected flight from late-aircraft cascade

Goal

Flight Cancellations cost everyone

$8B+

Annual revenue lost by U.S. airlines

~2-5%

of all flights cancelled (up to 5.6% for worst carriers)

25 min

Average delay per affected flight from late-aircraft cascade

Build a scalable big data pipeline that predicts which flights are at risk of cancellation - *before they are cancelled* - so airlines can act early

Dataset

flight delay & cancellation (kaggle)

Dataset

flight delay & cancellation (kaggle)

3 M	29.4 M	32	614 MB
flight records sampled	total flights in source	original columns	raw .csv size

Dataset

flight delay & cancellation (kaggle)

3 M	29.4 M	32	614 MB
flight records sampled	total flights in source	original columns	raw .csv size

Flight Info	Route	Times & Duration	Delays & Status	Delay Causes
FL_DATE	ORIGIN	CRS_DEP_TIME	DEP_DELAY	DELAY_DUE_CARRIER
FL_NUMBER	DEST	DEP_TIME	ARR_DELAY	DELAY_DUE_WEATHER
AIRLINE_CODE	DISTANCE	CRS_ARR_TIME	CANCELLED	DELAY_DUE_NAS
DOT_CODE	...	ARR_TIME	CANCELLATION_CODE	...
...		TAXI_IN	DIVERTED	
		...	...	

Dataset

flight delay & cancellation (kaggle)

3 M	29.4 M	32	614 MB
flight records sampled	total flights in source	original columns	raw .csv size

Flight Info	Route	Times & Duration	Delays & Status	Delay Causes
FL_DATE	ORIGIN	CRS_DEP_TIME	DEP_DELAY	DELAY_DUE_CARRIER
FL_NUMBER	DEST	DEP_TIME	ARR_DELAY	DELAY_DUE_WEATHER
AIRLINE_CODE	DISTANCE	CRS_ARR_TIME	CANCELLED	DELAY_DUE_NAS
DOT_CODE	...	ARR_TIME	CANCELLATION_CODE	...
...		TAXI_IN	DIVERTED	
		...	...	

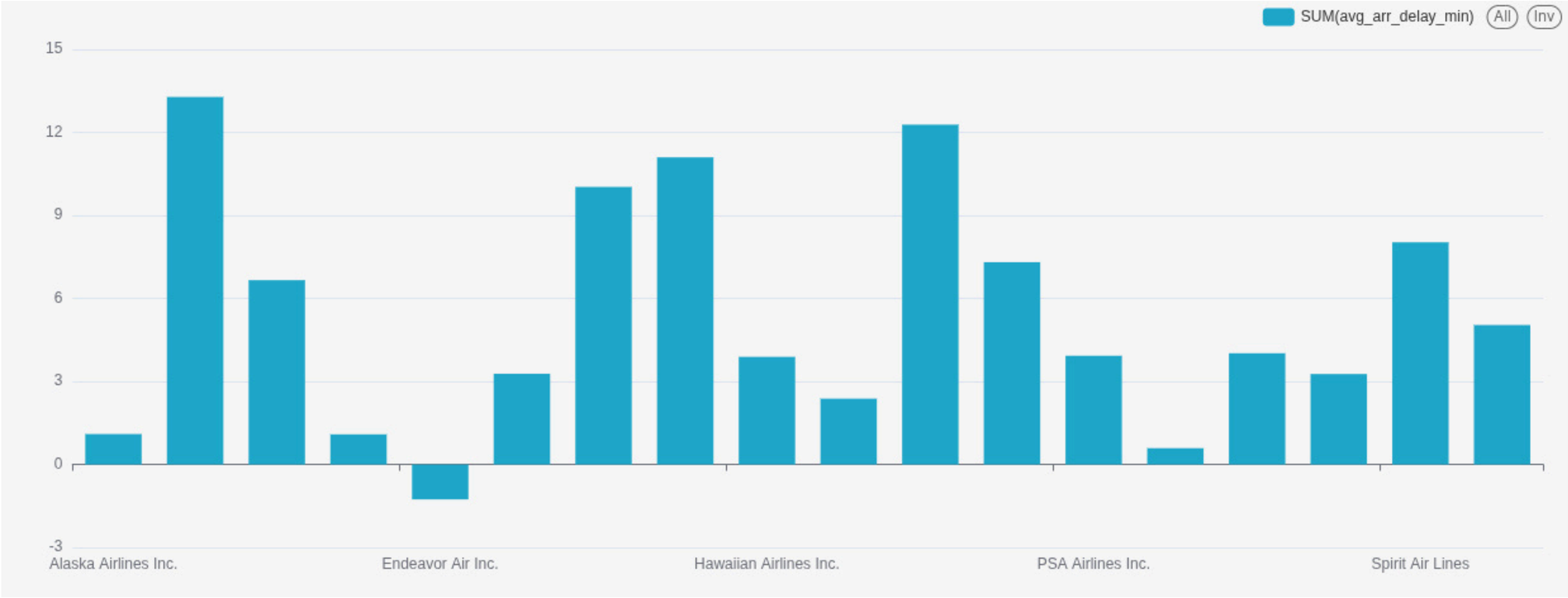
Coverage: January 2019 - August 2023

Data analysis

- **5 queries** → Hive result tables → CSV → Superset charts

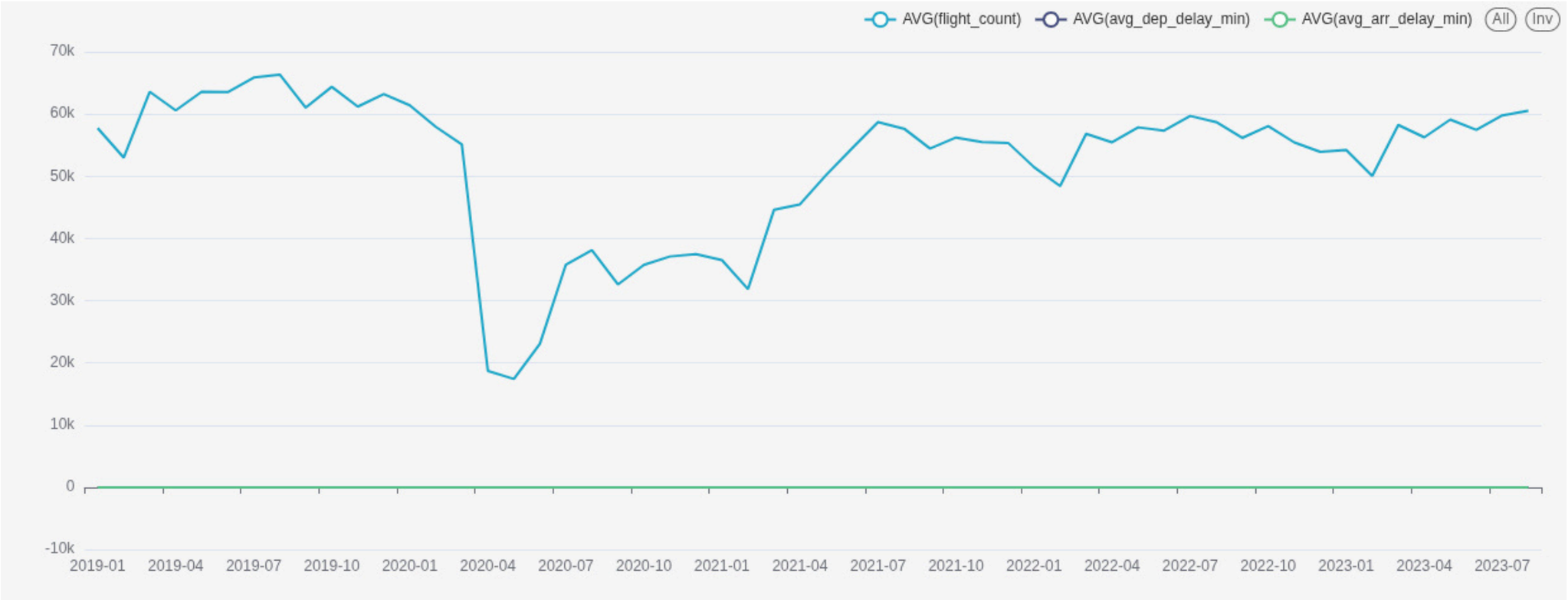
Data analysis

Avg arrival delay by airline



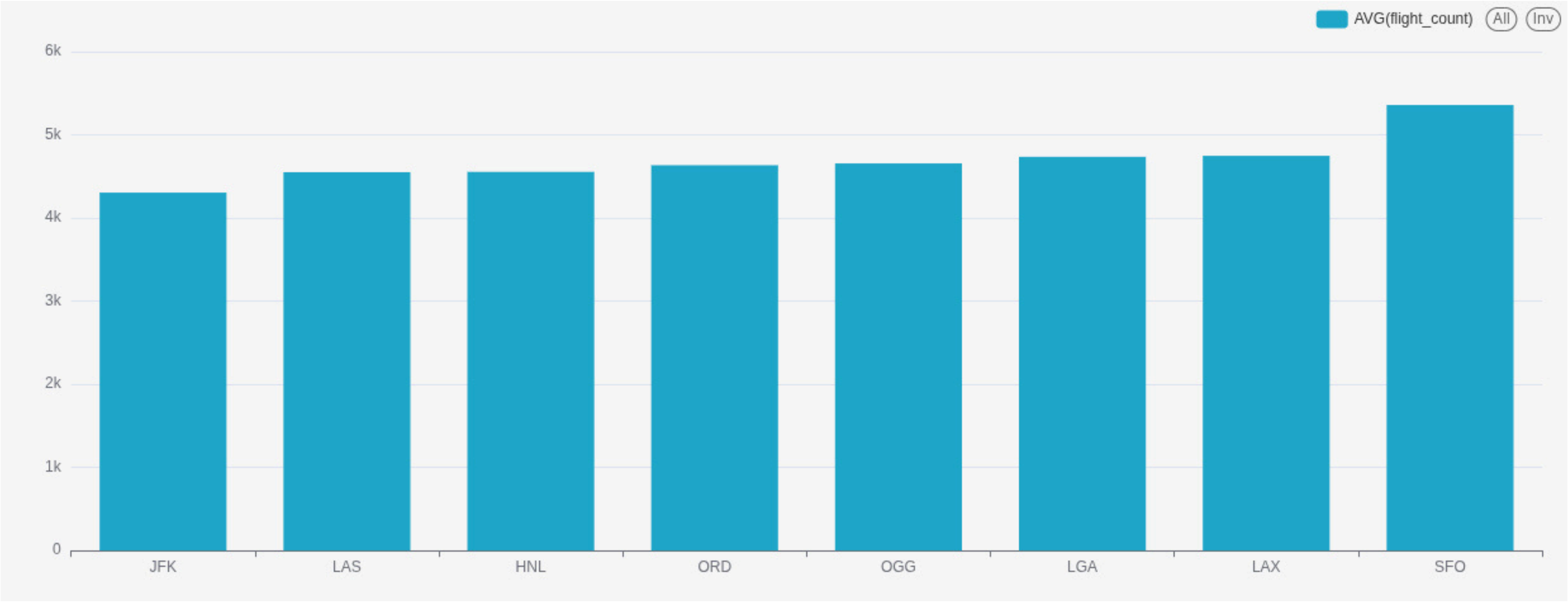
Data analysis

Monthly volume & delays



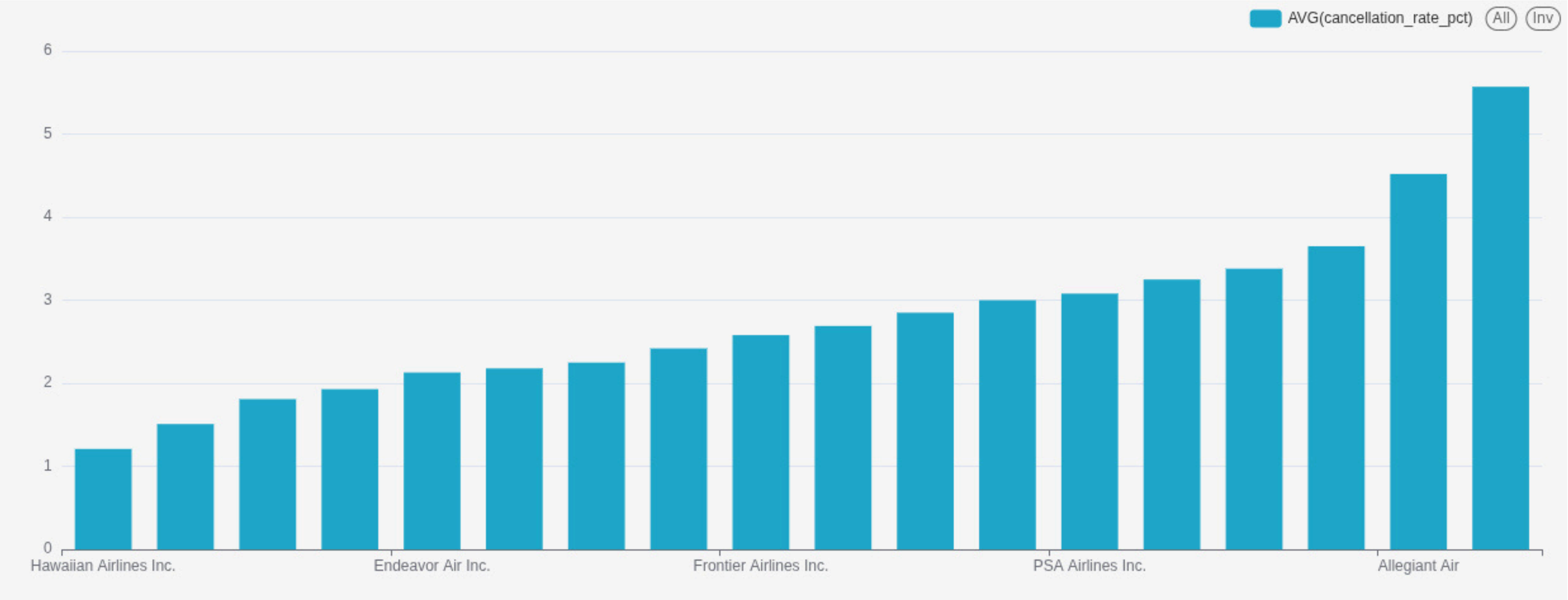
Data analysis

Top 10 busiest routes



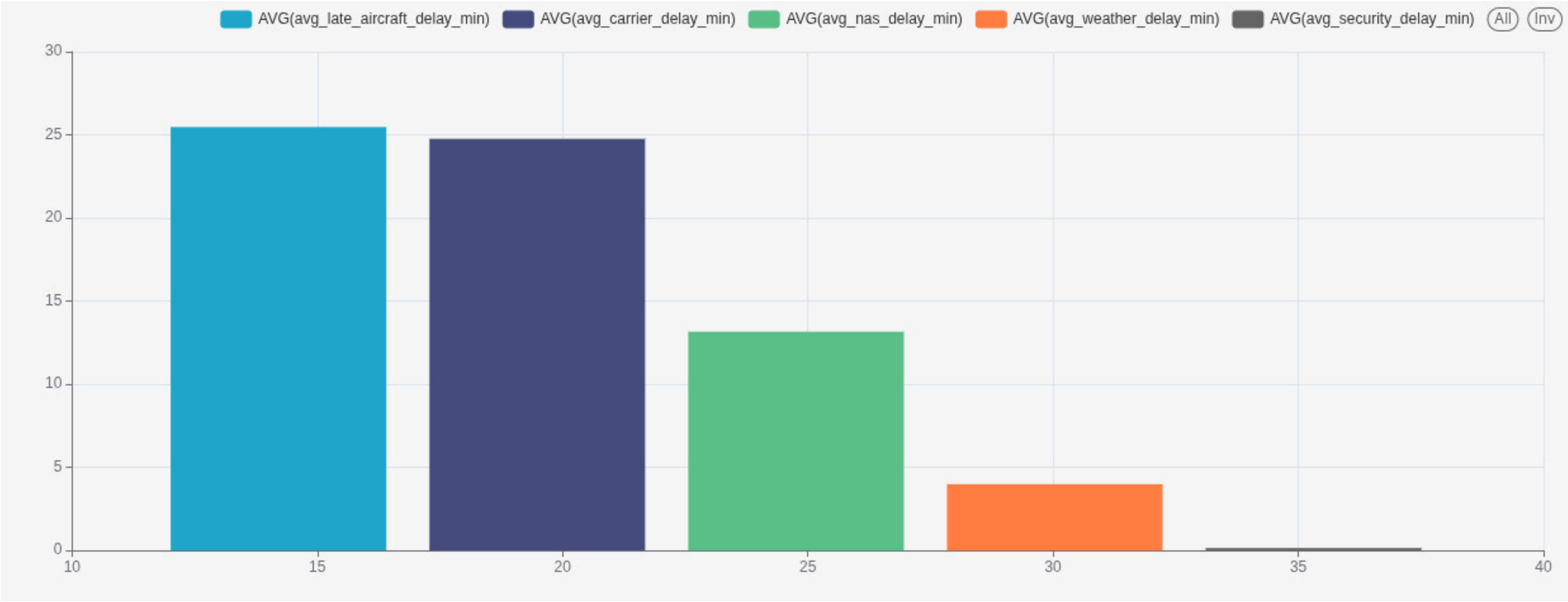
Data analysis

Cancellation rate by airline

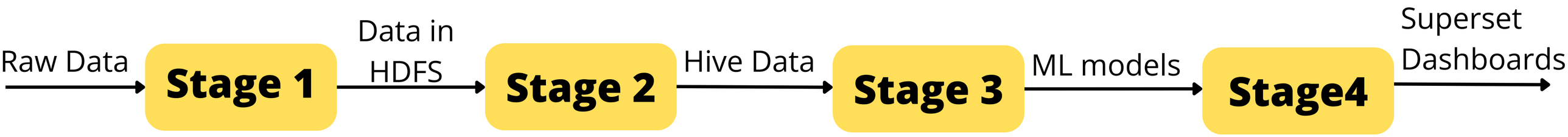


Data analysis

Avg delay by cause



Stages


Raw Data
Stage 1
Data in HDFS
Stage 2
Hive Data
Stage 3
ML models
Stage4
Superset Dashboards

Stage 1

Data Ingestion & Preprocessing

Stage 1

Data Ingestion & Preprocessing

Part 1: Preprocessing & PostgreSQL

- Raw CSV (614 MB) downloaded from Kaggle via kagglehub
- *Time columns:* HHMM integer → TIME; *Distance:* miles → km
- Deduplication on natural key
- Split into 3 normalized tables(airlines, airports, flights) → loaded into PostgreSQL

Stage 1

Data Ingestion & Preprocessing

Part 1: Preprocessing & PostgreSQL

- Raw CSV (614 MB) downloaded from Kaggle via kagglehub
- *Time columns:* HHMM integer → TIME; *Distance:* miles → km
- Deduplication on natural key
- Split into 3 normalized tables(airlines, airports, flights) → loaded into PostgreSQL

Part 2: HDFS Ingestion via Apache Sqoop

Exported all 3 tables from PostgreSQL → HDFS using Sqoop
Format: Parquet + Snappy (columnar + compressed)

Stage 1

Data Ingestion & Preprocessing

Part 1: Preprocessing & PostgreSQL

- Raw CSV (614 MB) downloaded from Kaggle via kagglehub
- *Time columns:* HHMM integer → TIME; *Distance:* miles → km
- Deduplication on natural key
- Split into 3 normalized tables(airlines, airports, flights) → loaded into PostgreSQL

Part 2: HDFS Ingestion via Apache Sqoop

Exported all 3 tables from PostgreSQL → HDFS using Sqoop
Format: Parquet + Snappy (columnar + compressed)

Output: 3 Parquet tables in HDFS, ready for Hive

Stage 2

Data Storage/Preparation & EDA

Stage 2

Data Storage/Preparation & EDA

Part 1: Hive Warehouse Design

- *Database team14_projectdb at /user/team14/project/hive/warehouse*
- 3 EXTERNAL tables on top of Stage 1 HDFS data: airlines, airports, flights
- Optimized fact table flights_part:
- Original flights dropped after migration → only optimized table remains

Stage 2

Data Storage/Preparation & EDA

Part 1: Hive Warehouse Design

- *Database team14_projectdb at /user/team14/project/hive/warehouse*
- 3 EXTERNAL tables on top of Stage 1 HDFS data: airlines, airports, flights
- Optimized fact table flights_part:
- Original flights dropped after migration → only optimized table remains

Part 2: Exploratory Data Analysis

- 5 analytical queries:
- Q1 — average arrival delay by airline
- Q2 — monthly flight volume + delays
- Q3 — top 10 busiest origin → destination routes
- Q4 — cancellation rate by airline
- Q5 — average delay by cause (carrier / weather / NAS / security / late aircraft)

Stage 2

Data Storage/Preparation & EDA

Part 1: Hive Warehouse Design

- *Database team14_projectdb at /user/team14/project/hive/warehouse*
- 3 EXTERNAL tables on top of Stage 1 HDFS data: airlines, airports, flights
- Optimized fact table flights_part:
- Original flights dropped after migration → only optimized table remains

Part 2: Exploratory Data Analysis

Output: 3 Hive tables + 5 result tables

- 5 analytical queries:
- Q1 — average arrival delay by airline
- Q2 — monthly flight volume + delays
- Q3 — top 10 busiest origin → destination routes
- Q4 — cancellation rate by airline
- Q5 — average delay by cause (carrier / weather / NAS / security / late aircraft)

Stage 3

Predictive Data Analytics

Part 1: Feature Engineering & Dataset Build (Hive → Spark)

- Read data from Hive: team14_projectdb.flights_part (Parquet on HDFS)
- Label: cancelled → binary label (0/1)
- Time features from fl_date (epoch ms): month, day_of_week
- Numeric features: fl_number, distance_km, crs_dep_time, crs_arr_time, + calendar features
- Categorical features: airline_code, origin, dest (encoded per model)

Stage 3

Predictive Data Analytics

Part 1: Feature Engineering & Dataset Build (Hive → Spark)

- Read data from Hive: team14_projectdb.flights_part (Parquet on HDFS)
- Label: cancelled → binary label (0/1)
- Time features from fl_date (epoch ms): month, day_of_week
- Numeric features: fl_number, distance_km, crs_dep_time, crs_arr_time, + calendar features
- Categorical features: airline_code, origin, dest (encoded per model)

Part 2: Model Training, Tuning & Evaluation (YARN)

- Train 3 required models: RandomForest, LinearSVC (SVM), NaiveBayes
- 70/30 train/test split
- 3-fold Cross-Validation + 3×3×3 grid search (optimized for AUC-PR)
- Evaluate on test with AUC-PR and AUC-ROC (plus accuracy/F1)

Stage 3

Predictive Data Analytics

Part 1: Feature Engineering & Dataset Build (Hive → Spark)

- Read data from Hive: team14_projectdb.flights_part (Parquet on HDFS)
- Label: cancelled → binary label (0/1)
- Time features from fl_date (epoch ms): month, day_of_week
- Numeric features: fl_number, distance_km, crs_dep_time, crs_arr_time, + calendar features
- Categorical features: airline_code, origin, dest (encoded per model)

Part 2: Model Training, Tuning & Evaluation (YARN)

- Train 3 required models: RandomForest, LinearSVC (SVM), NaiveBayes
- 70/30 train/test split
- 3-fold Cross-Validation + 3×3×3 grid search (optimized for AUC-PR)
- Evaluate on test with AUC-PR and AUC-ROC (plus accuracy/F1)

Output

- Saved PipelineModels in HDFS: .../models/model1–model3
- Predictions + metrics in HDFS: .../output/model*_predictions, .../output/evaluation.csv
- Optional mirrored CSVs in repo output/ for submission

Stage 4

Dashboard overview & storyline

DATA (Postgres)

- Row counts
- Column types
- Flights sample

EDA (Hive q1–q5)

- Monthly volume & delays
- Avg arrival delay
- Top 10 routes
- Cancellation rate
- Avg delay by cause

ML (Hive ml_*)

- Feature catalog
- Hyperparameter grid
- Model metrics
- 3× prediction samples

- http://hadoop-03.uni.innopolis.ru:8808/superset/dashboard/218/

Stage 4

Dashboard overview & storyline

Tech stack

Postgres
↓
Hive (external)
↓
Spark on YARN
↓
Apache Superset

2 DB connections:

- team14sql (PG)
- team14hive (Hive)

14 charts total

Stage 4

Data — Postgres characteristics

Data — row counts (Postgres)

table_name	column_name	data_type
airlines	airline_code	character varying
airlines	airline_dot	character varying
airlines	airline_name	character varying
airlines	dot_code	integer

Data — column types (Postgres)

table_name	row_count
airlines	18
airports	380
flights	3000000

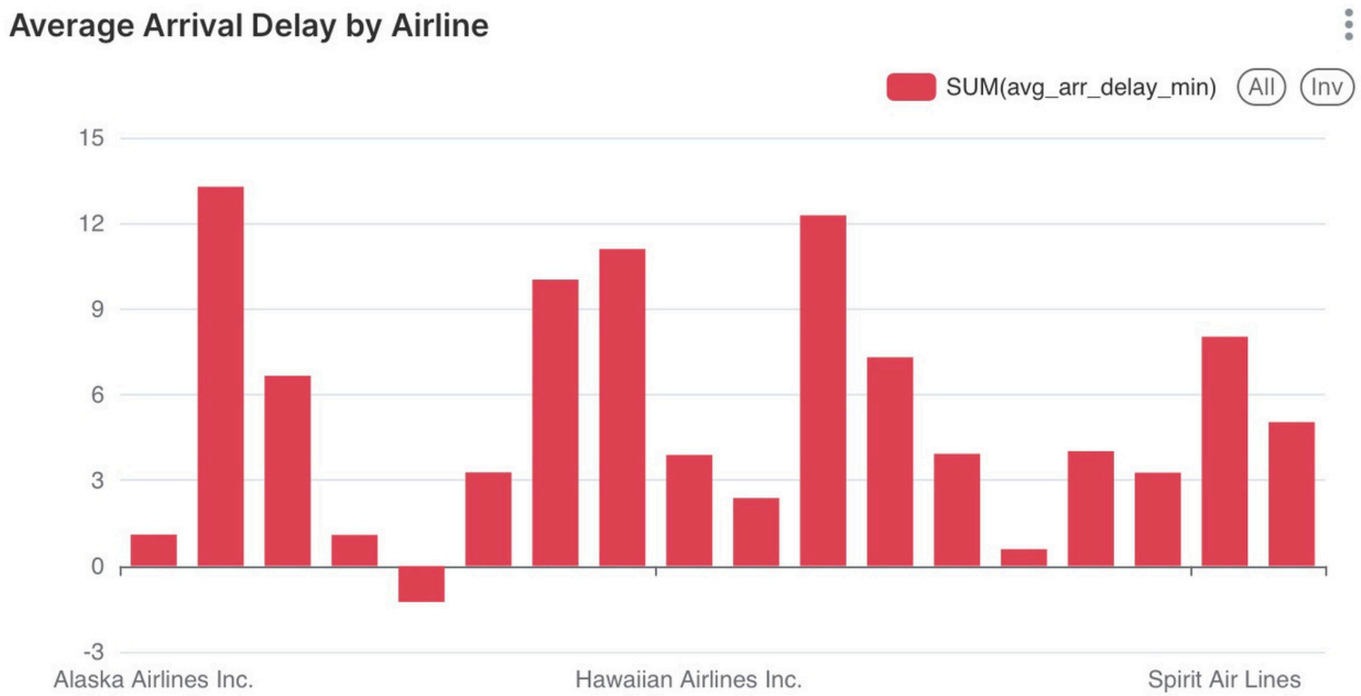
Postgres — flights sample (20)

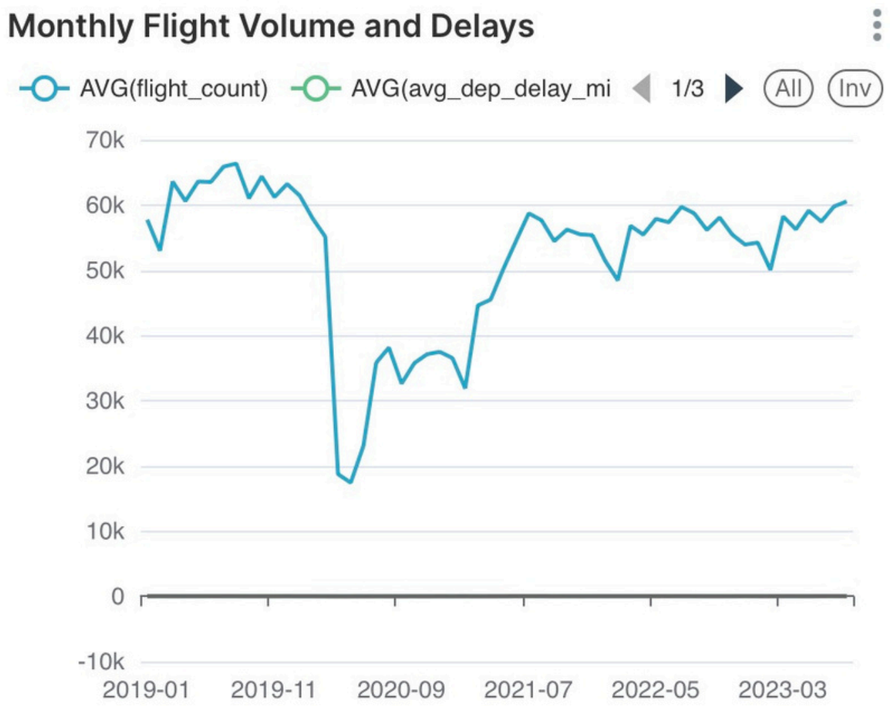
flight_id	airline_code	origin	dest	fl_date	cancelled	dep_delay	arr_delay
1	UA	FLL	EWR	2019-01-09	false	-4	-14
2	DL	MSP	SEA	2022-11-19	false	-6	-5
3	UA	DEN	MSP	2022-07-22	false	6	0
4	DL	MSP	SFO	2023-03-06	false	-1	24

- *Source: PostgreSQL · team14sql / public*

Stage 4

EDA — when & who (Hive q1, q2)

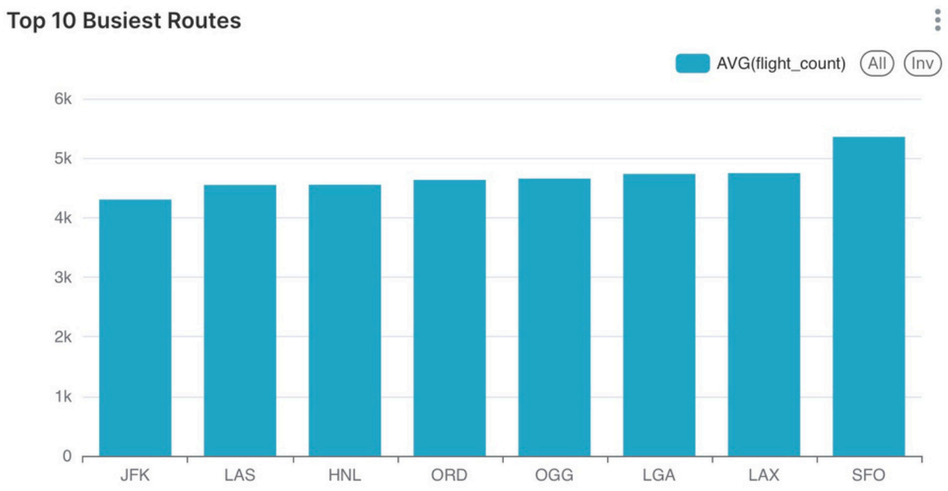


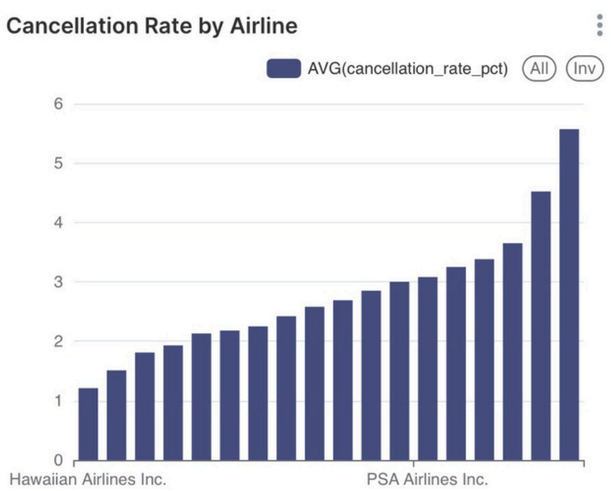


Q1: carriers spread from "on average early" to "consistently late".
Q2: 2020 collapse in volume and delays; recovery afterward read trends only with that break in mind.

Stage 4

EDA — where, cancellations, why (Hive q3, q4, q5)





- Busiest routes ≠ worst delays — hubs (e.g. SFO) move volume without catastrophe.
- Cancellation rate ≠ delay — different risk; carrier ranking changes.
- Among delayed flights: late aircraft and carrier dominate; weather / NAS / security less.

Stage 4

ML pipeline — features & tuning

ML — feature catalog

feature_name	feature_type	processing
airline_code	categorical	StringIndexer; LinearSVC/NaiveBayes use OneHotEncoder
crs_arr_time	numeric	scheduled arrival time (scaled HHMM-style integer)
crs_dep_time	numeric	scheduled departure time (scaled HHMM-style integer)
day_of_week	numeric	Spark dayofweek from flight timestamp

ML — hyperparameter search grids

model	parameter	search_space	best_selection_note
LinearSVC	aggregationDepth	values 2 3 4	aggregation depth for tree summaries
LinearSVC	regParam	values 0.01 0.05 0.2	CrossValidator maximizes areaUnderPR
LinearSVC	threshold	values 0.35 0.5 0.65	same evaluator (binary PR)
NaiveBayes	modelType	bernoulli and multinomial	binarized vs multinomial NB variants per Spark docs

- 3 models: RandomForest · LinearSVC · NaiveBayes (Spark MLlib)

Stage 4

Model comparison on test set

model	MAX(auc_roc)	MAX(auc_pr)
RandomForest	0.7221	0.0782
LinearSVC	0.5578	0.032
NaiveBayes	0.5219	0.0295

MAX(accuracy)	MAX(f1)
0.9739	0.9611
0.9739	0.9611
0.9735	0.9605

- Class imbalance → leading metric is PR AUC, not ROC.
  Best: Random Forest — best PR AUC and accuracy; F1 close across models.

Stage 4

Predictions & takeaways

ML — RF held-out predictions (sample)

label	prediction
0	0.0
0	1.0

ML — LinearSVC held-out predictions (sample)

label	prediction
0	0.0
1	0.0

ML — NaiveBayes held-out predictions (sample)

label	prediction
0	0.0
1	0.0

- Sample held-out predictions for all 3 models (RF / LinearSVC / NaiveBayes).
- The Best for the dashboard: Random Forest (PR-oriented).
- Next: regression on delay minutes for non-cancelled flights.

Challenges

- Ingestion of data in PostgreSQL when data has inappropriate format.
- Workaround class imbalance in ML.
- Coordinate the teamwork with GitHub.
- High-cardinality categoricals forced different pipelines to avoid memory blow-ups and invalid inputs for Bernoulli NB.

Results

- Built automated end-to-end Big Data pipeline.
- Identified carrier-specific delay performance patterns.
- Discovered main causes of flight delays.
- Evaluated distributed machine learning models.

Thanks for attention!